ZEROTH REVIEW

Design of a Frequency Synthesizer

A PLL-based CMOS frequency synthesizer, designed and simulated in Cadence Virtuoso

TEAM

Aaryan Dogra 23BEC1032

Yugeshwaran 23BEC1404

Viju Sanjai 23BEC1306

Project Guide **Dr. B. Bindu**

School of Electronics Engineering (SENSE)

Introduction

Context, applications, and where this project fits

Frequency synthesizers generate a range of precise, stable output frequencies from a single reference oscillator — a core building block of nearly every RF and digital system.

- Wireless transceivers & 5G/6G RF front-ends
- Radar and satellite communication systems
- Clock generation for digital / SoC systems
- Test & measurement instrumentation

Why Temperature Matters

PLL/VCO frequency drifts with temperature-dependent device parameters (Vt, mobility, capacitance) — causing loss of lock, spectral impurity, and timing error in real operating environments.

-40°C to +85°C

typical operating range

ppm-level

stability required

This project focuses on the design and Cadence-based simulation of a PLL frequency synthesizer with an integrated temperature-compensation scheme.

Problem Statement

The gap this project addresses

Conventional PLL-based frequency synthesizers exhibit significant output-frequency drift, phase-noise degradation, and loop-locking instability as ambient temperature varies — limiting their reliability in field-deployed RF and communication systems.

1

VCO Frequency Drift

Device parameter shift (Vt, mobility) with temperature detunes the VCO's free-running frequency.

2

Phase Noise Degradation

Temperature-dependent noise sources erode spectral purity away from nominal test conditions.

3

Loop Instability

Wide thermal swings can push the PLL out of its locking range, delaying or breaking lock.

[1] [2] [3]

Objectives

What this project sets out to achieve

1 Design a PLL-based synthesizer

Architect a CMOS PLL frequency synthesizer targeting a defined output band and reference frequency.

2 Integrate temperature compensation

Combine static and dynamic compensation techniques to stabilize output frequency across temperature.

3 Simulate & verify in Cadence

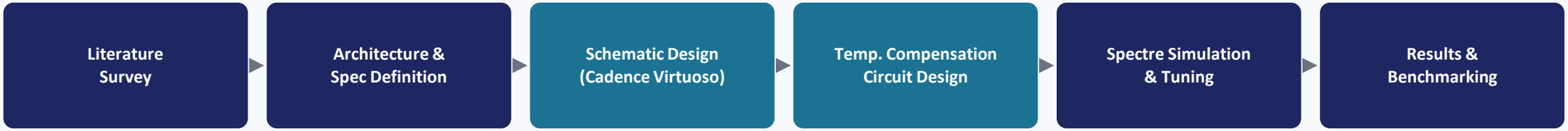
Validate lock time, phase noise, and frequency stability via Virtuoso / Spectre simulation.

4 Benchmark against literature

Compare achieved specs (ppm drift, phase noise, power) against published designs.

Proposed Methodology

From architecture definition to simulation-based verification



Design Flow Detail

- Reference oscillator → Phase-Frequency Detector (PFD) → Charge Pump → Loop Filter → VCO → Feedback Divider
- Temperature sensor output modulates a compensation code applied to the VCO control path (static + dynamic compensation)
- Simulate across process corners and –40°C to +85°C using Cadence Spectre; verify lock time, phase noise, and frequency error — entirely at schematic / testbench level, no physical layout

Reference Precedent

Fast-locking CMOS PLL implemented in Cadence Virtuoso GPDK 45nm — closest methodology template for this project, adapted here to a Virtuoso-only simulation flow.

Expected Outcomes

Target performance envelope

± 0.15 ppm	−40°C → 85°C	< 2 µs	< −80 dBc/Hz
frequency stability across the temperature range	compensated operating temperature range	PLL locking time target	phase noise @ 10 kHz offset

Deliverables

- Verified Cadence schematic and testbench for the compensated PLL synthesizer
- Simulation report: lock time, phase noise, and frequency-vs-temperature plots
- A publishable comparison against benchmarked prior work

Targets benchmarked against [1] [2] [3] [4]

Tools & Technologies

The design and verification stack

Virtuoso Schematic Editor

Schematic capture and design entry within the Cadence Virtuoso environment

Spectre Simulator

Core transient / AC / periodic-steady-state circuit simulation engine

Virtuoso ADE Explorer

Single-run simulation setup, analysis, and results plotting

Virtuoso ADE Assembler

Multi-corner and Monte Carlo sweep environment across PVT

Spectre RF (PSS / PNoise)

Periodic steady-state and phase-noise analysis for the VCO/PLL

Foundry CMOS PDK

Process design kit (180nm / 45nm) loaded into Virtuoso for device models

Project Timeline

Four scheduled reviews across the project duration

Review 0

Literature survey, problem definition, architecture & spec finalization

Review 1

Schematic design of PLL blocks (PFD, CP, LF, VCO) in Cadence Virtuoso

Review 2

Temperature-compensation circuit integration & Spectre simulation across PVT

Review 3

Final verification, results consolidation, and report / paper draft

Final submission / publication draft follows Review 3

References

Key papers informing this design

[1] **Low Power Fast Locking PLL Frequency Synthesizer with Temperature Compensation.**

Proc. 3rd Int. Conf. on Electronic, Mechanical, Automation and Materials Engineering (EAME), Atlantis Press, 2018.

[2] **A 12GHz 67% Tuning Range, 0.37ps RJrms PLL with LC-VCO Temperature Compensation Scheme in 0.13µm CMOS.**

IEEE conference publication, 2014.

[3] **Design of an Integrated CMOS PLL Frequency Synthesizer.**

IEEE Conference Publication, IEEE Xplore.

[4] **Temperature and Power Supply Compensated CMOS Clock Circuit Based on Ring Oscillator.**

Electronics, MDPI, vol. 12, no. 3, 2023.

[5] **Phase-Locked Loop (PLL)-Based Frequency Synthesizer for Digital Systems Driving.**

Cadence Virtuoso GPDK 45nm CMOS implementation, ResearchGate.